Title page

Beyond Directional Accuracy: A Verified Walk-Forward Framework for Machine-Learning Cryptocurrency Trading

Ratnachand Kancharla[a,*] and Sai Srinivas Panthangi[b]

[a] NMIMS Hyderabad Campus, Hyderabad, Telangana, India
[b] Malla Reddy University, Hyderabad, Telangana 500100, India

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*** Corresponding author**

Ratnachand Kancharla
NMIMS Hyderabad Campus, Hyderabad, Telangana, India
Email: ratnachandkancharla@gmail.com
ORCID: https://orcid.org/0009-0003-5634-8209

Author details

Ratnachand Kancharla (corresponding author), NMIMS Hyderabad Campus, Hyderabad, Telangana, India. Email: ratnachandkancharla@gmail.com. ORCID: 0009-0003-5634-8209.

Sai Srinivas Panthangi, Malla Reddy University, Hyderabad, Telangana 500100, India. Email: saisrinivaspanthangi09@gmail.com. ORCID: 0009-0009-2363-4667.

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CRediT authorship contribution statement

Ratnachand Kancharla: Conceptualization, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Writing – review & editing.

Sai Srinivas Panthangi: Software, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

The source code, the three input files, the dated out-of-sample forecasts, the trade ledger and all machine-readable results are available at https://github.com/Ratnachand04/FINBIN. The market data are public Binance kline archives. The full analysis is rebuilt by the command python scripts/reproduce_jfds.py.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work the authors used OpenAI Codex and Anthropic Claude to help with parts of the analysis code and tests, the search for related literature, and the writing and checking of the manuscript. No data and no reported number was produced by a language model. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. The full statement is given at the end of the manuscript.

Ethics statement

The study uses only public market data and software written by the authors. It does not involve human participants, personal data or animals, so ethics approval and informed consent were not needed.